Aaron Lin

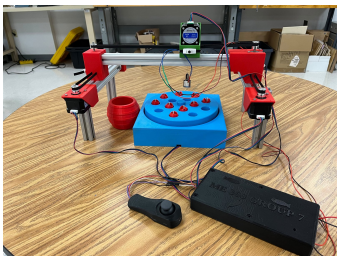
Mechanical design, analysis, prototyping, and testing

B.S. Mechanical Engineering, Aerospace Concentration · Boston University · May 2028
Materials Science & Engineering minor

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Boston, Massachusetts

I am a mechanical engineering junior seeking a **Summer 2027 internship** in mechanical design, product development, or manufacturing. My recent work spans electromechanical systems, aerodynamic analysis, and mechanical hardware from CAD through fabrication and testing.

Selected Engineering Work



2.5-DOF Cartesian Fishing Game

My Role: CAD lead · Completed

Belt-driven fishing game with joystick-controlled X-Y motion, motorized Z pickup, magnetic hook, and rotating target board.

Demo Complete
X-Y-Z control and all three game rounds worked

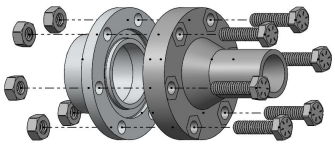


ARGO Aerodynamic Fairing

My Role: Project Engineer · In progress

Installed-geometry fairing concepts for an external downcomer and wiring at a Mach 2.81 design condition.

CDR Complete
September 2026; comparative analysis in progress



Converging-Diverging Rocket Nozzle

My Role: Mechanical Lead · Completed

Flow sizing, CAD, drawings, pressure analysis, resin printing, and static-fire integration.

2.12
minimum factor of safety in the pressure-only model

Experience and Capabilities

Mechanical Engineering Intern, Culturon
Sydney, Australia · February–June 2026
Ran optical emission spectroscopy experiments and built Python tools with pandas, SciPy, and Astroquery to process plasma spectra and compare candidate lines with the NIST Atomic Spectra Database.

Project Engineer, Boston University Rocket Propulsion Group
September 2024–present
Designed and fabricated a resin rocket nozzle; now comparing aerodynamic fairing concepts using Ansys Fluent, OpenRocket, and first-principles calculations.

Design and Analysis
SolidWorks · Fusion360 · Fusion CAM · Ansys Fluent · OpenRocket · FEA · CFD · GD&T fundamentals

Programming and Data
Python · pandas · SciPy · Astroquery · MATLAB · C++ (Arduino) · Java · GCode

Fabrication and Test
Resin and FDM printing · CNC Mill/Lathe · Soldering · Sensors · Experimental Testing · Welding · Geometric Dimensioning and Tolerancing

More drawings, reports, test media, and complete case studies: aaronlin.space.

2.5-DOF Cartesian Fishing Game

Boston University · Electromechanical Design · Team Project · September–October 2026 · Completed

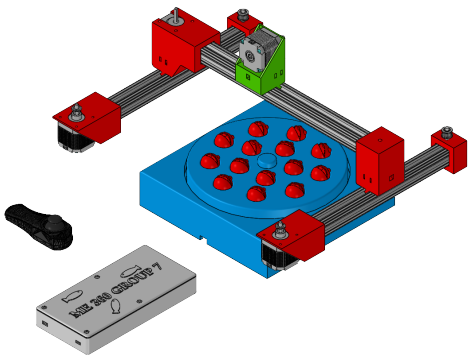
My Role: CAD lead

My contribution: CAD and mechanical layout for the belt-driven gantry, Z pickup, game board, joystick, and controller packaging

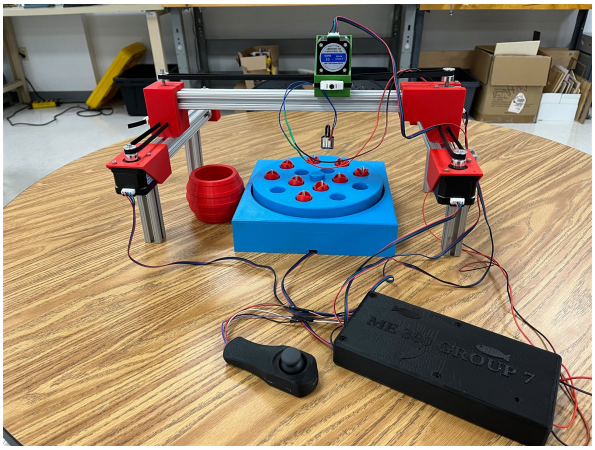
Team outcome: Team fabricated, integrated, and demonstrated the completed electromechanical game

Evidence status: X-Y-Z control, magnetic pickup, rotating board, and all three rounds worked; travel bounds remain uncalibrated

Our team completed a joystick-controlled fishing game around two synchronized X rails, one Y rail, and a constrained motorized Z motion. An electromagnet collects pinned game pieces while a separate stepper motor drives the rotating target board.



Assembly CAD used to coordinate motion hardware, game elements, controls, and packaging.



Completed prototype used for the classroom demonstration.

Design Decisions

Two synchronized X rails support the crossbeam and limit canting under the Y-axis and end-effector load. Belt-driven stages on 80/20 extrusion provide the required work area while keeping the structure modular. The Z motor locks X-Y motion during the controlled pickup-and-return sequence.

Integration and Result

The team corrected an initially reversed X control in software before the classroom demonstration. Joystick-controlled X-Y motion, the automatic Z dip, magnetic pickup, rotating board, and all three programmed rounds then operated together.

Next steps: Calibrate the X-Y travel bounds, add homing or software limits, and provide a visible indicator when the electromagnet is energized.

Ownership note: I led CAD and mechanical layout. Fabrication, controls, integration, and testing were team work.

Methods: SolidWorks · FDM Printing · 80/20 Extrusion · Timing Belts · Arduino · Prototype Testing

Full case study, control code, and test media

ARGO Aerodynamic Fairing

Boston University Rocket Propulsion Group · Project Engineer · July 2026–present · In Progress

My Role: Project Engineer

My contribution: Baseline and candidate geometry, stagnation calculations, material screening, and CFD preparation

Team outcome: Conceptual Design Review completed

Evidence status: Current milestone documented; comparative simulation results pending



Preliminary opaque aerocover geometry prepared for installed-configuration analysis.



Transparent view showing the protected routing and the aerocover's relationship to the installed hardware.

Mach 2.81

design condition for the current aerodynamic comparison

Design Problem

ARGO routes a downcomer and wiring along the rocket exterior, creating aerodynamic, thermal, attachment, and packaging challenges. The fairing must protect those components while limiting drag and remaining compatible with the installed geometry.

My Work

I am comparing the previous Icarus aerocover with candidate leading edges, midsections, and trailing-edge or boattail concepts using Ansys Fluent, OpenRocket, and first-principles calculations. I translate the installed packaging constraints into consistent baseline and candidate geometries and document the assumptions used in each comparison.

CDR Completed

September 3, 2026; comparative simulation results remain pending

Completed Milestone

Work to date includes stagnation temperature and pressure calculations, early material screening, and preparation of candidate CFD geometry.

Comparison Controls

A body-mounted fairing is only a partial revolution, so a fully revolved nose cone is not a direct substitute. Baseline and candidates must use matched installed geometry, boundary conditions, reference area, and reporting methods.

Next Analysis

Before Preliminary Design Review, I will document the mesh and compare one checked performance metric across matched cases.

Limits: This preliminary geometry is not a completed performance claim. No drag reduction, pressure result, or selected geometry is reported until the simulations are complete and checked for mesh and setup consistency.

Methods: Ansys Fluent · OpenRocket · Supersonic Flow · First-Principles Calculations

Full case study and current design status

Converging-Diverging Rocket Nozzle

Boston University Rocket Propulsion Group · Fall 2024 · Completed

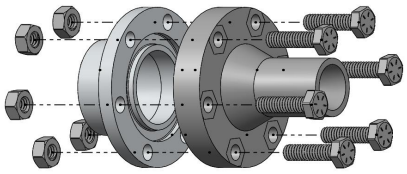
My Role: Individual mechanical lead

My contribution: Flow sizing, CAD, drawings, pressure analysis, fabrication, and integration

Team outcome: Integrated with the chamber and ignition system for static-fire testing

Evidence status: Pressure model and build artifacts documented; performance not measured

I owned the nozzle's mechanical design and fabrication: flow-path sizing, CAD and drawings, resin printing, and static-fire integration. The design used an 80 psi chamber-pressure requirement, 3250 K chamber temperature, 13 g/s mass flow, and a 2.1 psia target exit pressure.



Nozzle, six-bolt adapter, and fasteners.



Static-fire integration recording.

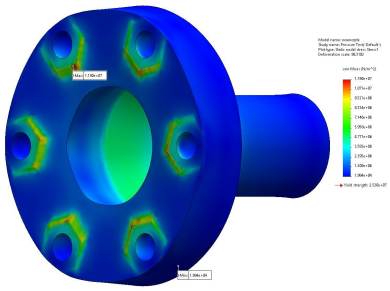
Design Decisions

I used isentropic-flow relations to size the throat and exit, kept the assembly within a 1.05 in diameter limit, and modeled the nozzle and adapter in SolidWorks. Resin printing supported the internal geometry and schedule before integration with the chamber and seals.

Pressure-Analysis Evidence

The SolidWorks Simulation model applied approximately 46.3 psi at the throat and 80 psi in the chamber. It reported a maximum von Mises stress of 11.90 MPa against an assumed 25.20 MPa yield strength.

Limits: The 2.12 factor of safety applies only to the pressure model and its material and boundary assumptions. The hot-fire recording demonstrates system integration, not quantified nozzle performance.



11.90 MPa maximum stress versus 25.20 MPa assumed yield; deformation exaggerated.

6.521 exit-to-throat area ratio

2.12 minimum factor of safety · pressure-only model

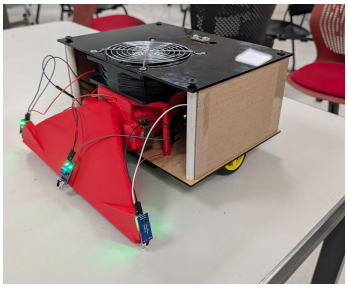
Next Validation. Add thermal properties, heat transfer, mesh convergence, instrumentation, and a documented post-fire inspection.

Methods: SolidWorks Simulation · Pressure FEA · Compressible Flow · Resin Printing

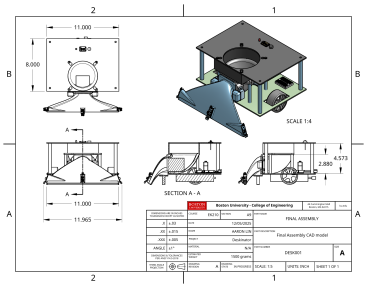
Full case study and hot-fire videos · Assembly drawing (PDF)

Selected Build, Test, and Analysis Work

Earlier Projects and Industry Experience · Supporting Evidence



Prototype used for presentation and pickup testing.



Assembly drawing showing packaging and service access.

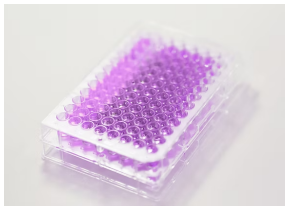
The Deskinator

Boston University · Introduction to Engineering Design · Team of Four · Fall 2025

I wrote the Arduino C++ navigation firmware for S-curve pathing, IR-triggered edge avoidance, and gesture activation, and helped assemble and troubleshoot the mechanical and electrical systems. **76% pickup rate** in one documented final trial (72 of 95 rice grains); edge avoidance and gesture activation were demonstrated but not fully repeatable.

Mechanical Engineering Intern, Culturon

Sydney, Australia · February–June 2026

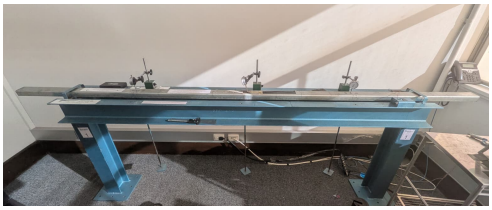


Public Culturon PAC microtitre plate; product image from culturon.com.au.

I ran optical emission spectroscopy experiments and built Python tools with pandas, SciPy, and Astroquery to process plasma spectra and compare candidate lines with the NIST Atomic Spectra Database.

Bending of a Steel Channel Section

University of Sydney · Structural Mechanics · Spring 2026



Steel-channel bending setup with three dial gauges and symmetric loading points.

At the largest load, the strain gauge measured **71.44 microstrain** and the curvature calculation predicted **72.63 microstrain**, a **1.63% difference**. The measurements were gathered with classmates and are presented as team evidence.

Additional Work Online

Temperature Monitor	Individual Arduino build with sensing, dual-unit display, and high and low-temperature alerts.
Diffusion Analysis	Analyzed dye concentration in time-series images to fit Fick's 2nd law coefficients with MATLAB.
Acrylic Truss	The final MATLAB model predicted 46.59 oz; the physical build met the 16 oz requirement.
SolidWorks Exercises	Parts, assemblies, section views, exploded views, drawings, bills of materials, and motion studies.

Full case studies, drawings, reports, and test media: aaronlin.space.

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